INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	WA facility WA_way_499403699 -- station KEPH, America/Los_Angeles
Committed pair	OSM 283051478 -> 499403699 OSM way 283051478 / OSM way 499403699
Facade gap	99.3 m between the two halls
Weather record	42,965 real hourly records from KEPH
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.4420 C at 250 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-04-18 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 1 of 24 hours with 1 mode change. The reactive incumbent took 11 hours -- 10 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	1 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	11 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	8.139	18.0	7.780	1.187
01	mechanical	yes	switch budget	8.691	18.0	8.890	1.123
02	mechanical	yes	switch budget	7.573	18.0	6.670	1.089
03	mechanical	yes	switch budget	8.643	18.0	8.890	1.065
04	mechanical	yes	switch budget	7.294	18.0	5.000	1.063

05	mechanical	yes	switch budget	5.887	18.0	4.440	0.941
06	mechanical	yes	switch budget	8.893	18.0	6.299	1.326
07	mechanical	yes	switch budget	9.777	18.0	9.440	1.817
08	mechanical	yes	switch budget	12.661	18.0	13.127	1.951
09	mechanical	yes	switch budget	15.028	18.0	15.560	1.850
10	mechanical	no	dry-bulb	18.413	18.0	18.890	1.851
11	mechanical	no	dry-bulb	20.838	18.0	21.299	1.604
12	mechanical	no	dry-bulb	23.176	18.0	24.440	1.397
13	mechanical	no	dry-bulb	25.628	18.0	27.527	1.230
14	mechanical	no	dry-bulb	26.722	18.0	28.416	1.068
15	mechanical	no	dry-bulb	28.416	18.0	28.890	0.994
16	mechanical	no	dry-bulb	28.687	18.0	28.330	0.963
17	mechanical	no	dry-bulb	28.413	18.0	27.220	1.240
18	mechanical	no	dry-bulb	26.997	18.0	25.000	1.189
19	mechanical	no	dry-bulb	25.347	18.0	22.780	1.367
20	mechanical	no	dry-bulb	23.919	18.0	20.560	1.887
21	mechanical	no	dry-bulb	22.250	18.0	19.440	1.711
22	mechanical	no	dry-bulb	20.312	18.0	17.780	1.382
23	FREE	yes	-	16.703	18.0	12.220	1.268

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.139 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.691 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.573 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.643 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.294 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.887 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.893 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.777 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.661 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.028 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.413 C against a 18.0 C limit, so it fails by 0.413 C. It would take a limit 0.413 C higher, or a bound 0.413 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.838 C against a 18.0 C limit, so it fails by 2.838 C. It would take a limit 2.838 C higher, or a bound 2.838 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.176 C against a 18.0 C limit, so it fails by 5.176 C. It would take a limit 5.176 C higher, or a bound 5.176 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.628 C against a 18.0 C limit, so it fails by 7.628 C. It would take a limit 7.628 C higher, or a bound 7.628 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.722 C against a 18.0 C limit, so it fails by 8.722 C. It would take a limit 8.722 C higher, or a bound 8.722 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.416 C against a 18.0 C limit, so it fails by 10.416 C. It would take a limit 10.416 C higher, or a bound 10.416 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.687 C against a 18.0 C limit, so it fails by 10.687 C. It would take a limit 10.687 C higher, or a bound 10.687 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.413 C against a 18.0 C limit, so it fails by 10.413 C. It would take a limit 10.413 C higher, or a bound 10.413 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.997 C against a 18.0 C limit, so it fails by 8.997 C. It would take a limit 8.997 C higher, or a bound 8.997 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.347 C against a 18.0 C limit, so it fails by 7.347 C. It would take a limit 7.347 C higher, or a bound 7.347 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.919 C against a 18.0 C limit, so it fails by 5.919 C. It would take a limit 5.919 C higher, or a bound 5.919 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.250 C against a 18.0 C limit, so it fails by 4.250 C. It would take a limit 4.250 C higher, or a bound 4.250 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.312 C against a 18.0 C limit, so it fails by 2.312 C. It would take a limit 2.312 C higher, or a bound 2.312 C tighter, to change this hour.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.702 C, which is 1.298 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.268 C, and the margin is measured, not chosen: 1.231 C of group-conditional forecast error for this hour of day, plus 0.0375 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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